

VPC CHALLENGE

Task 1: Transit Gateway Setup -

Establishing Inter-VPC Connectivity Using AWS Transit Gateway

Objective:

Design and deploy a scalable network architecture using AWS Transit Gateway to enable secure communication between multiple VPCs.

Architecture Example:

VPC-A: 192.168.1.0/24

VPC-B: 192.168.2.0/24

VPC-C: 192.168.3.0/24

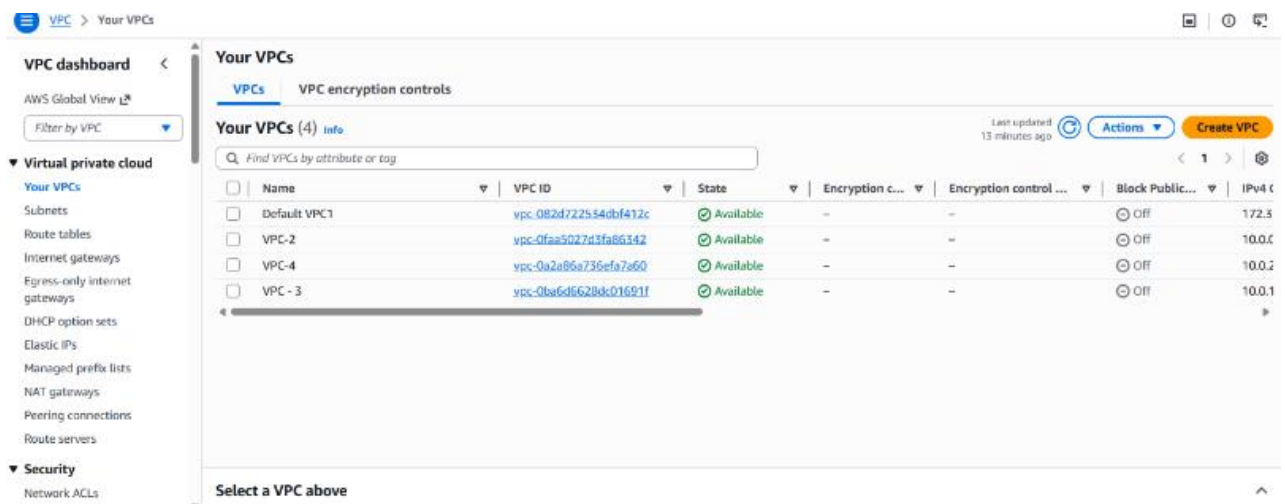
Transit Gateway: Central Hub

Step 1: Create VPC-1 (Default)

Go to AWS Console → VPC → Your VPCs → Create VPC

Name: VPC-1

CIDR: 192.168.1.0/24

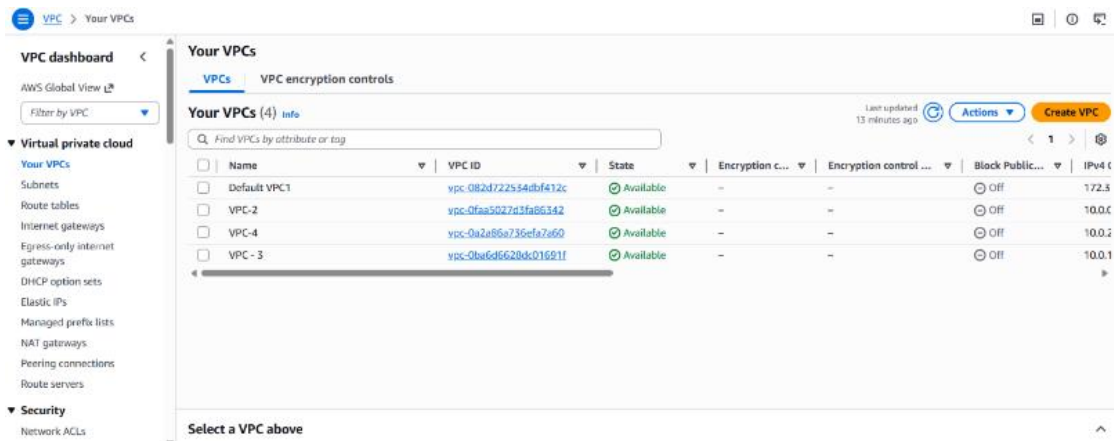


Step 2: Create VPC-2

Go to AWS Console → VPC → Your VPCs → Create VPC

Name: VPC-2

CIDR: 10.0.0.0/28

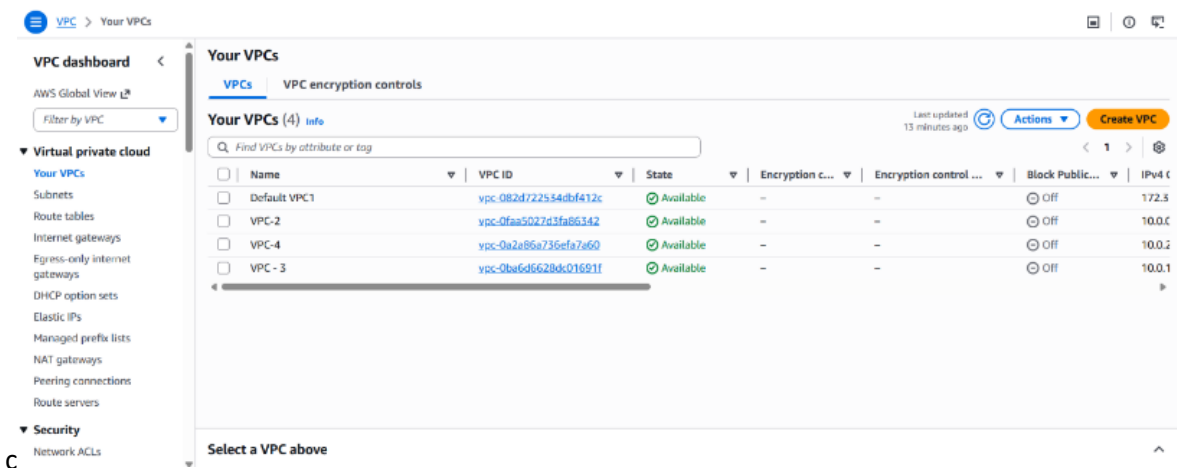


Step 3: Create VPC-3

Go to AWS Console → VPC → Your VPCs → Create VPC

Name: VPC-3

CIDR: 10.0.1.0/28

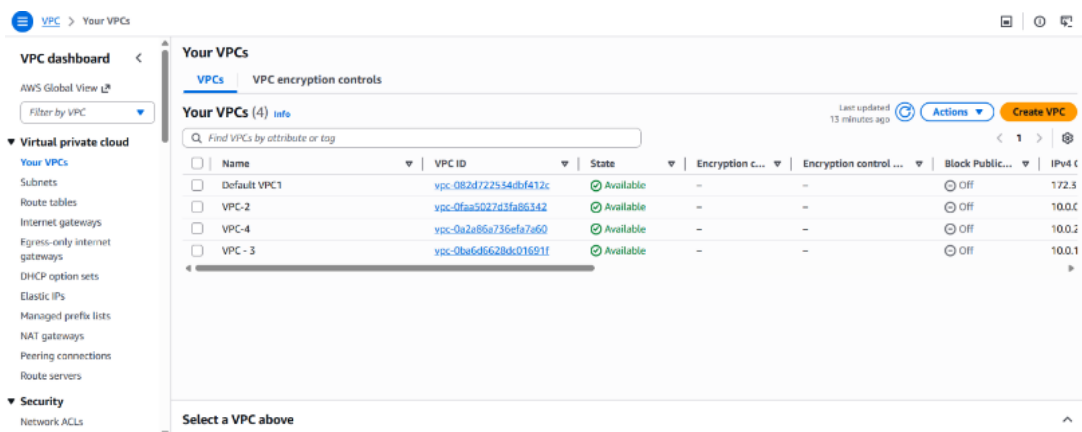


Step 4: Create VPC-4

Go to AWS Console → VPC → Your VPCs → Create VPC

Name: VPC-4

CIDR: 10.0.2.0/28

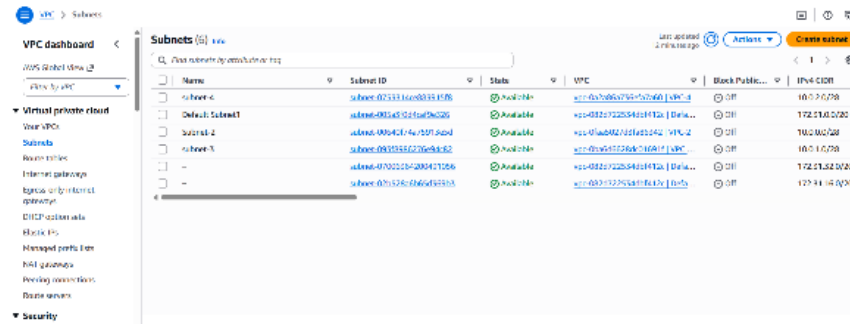


Step 5: Create Subnet in VPC-1 (Default)

Go to Subnets → Create Subnet

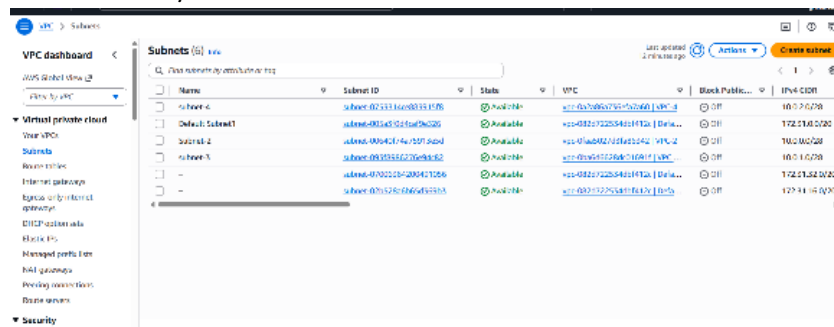
Select VPC-1

CIDR: 192.168.1.0/25



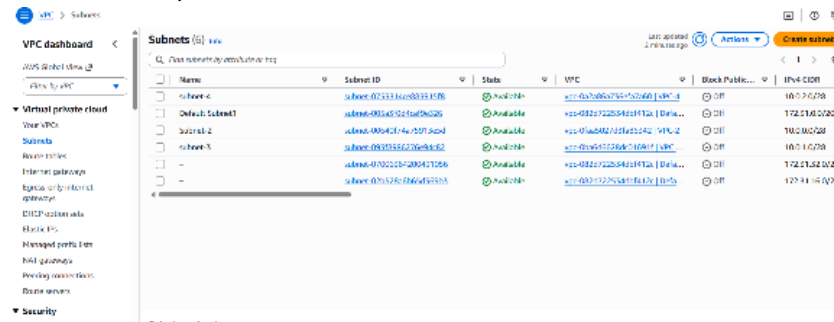
Step 6: Create Subnet in VPC-2

CIDR: 10.0.0.0/28



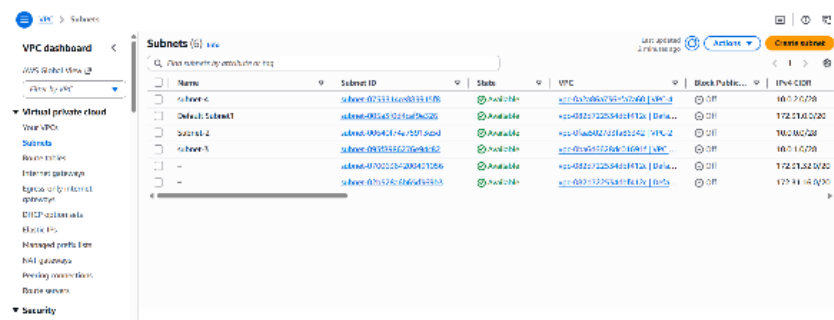
Step 7: Create Subnet in VPC-3

CIDR: 10.0.1.0/28



Step 8: Create Subnet in VPC-4

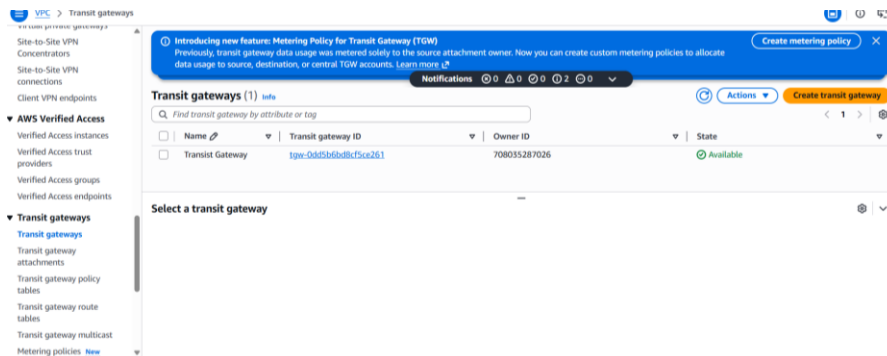
CIDR: 10.0.2.0/28



Step 9 : Create Transit Gateway

Go to VPC → Transit Gateways → Create Transit Gateway

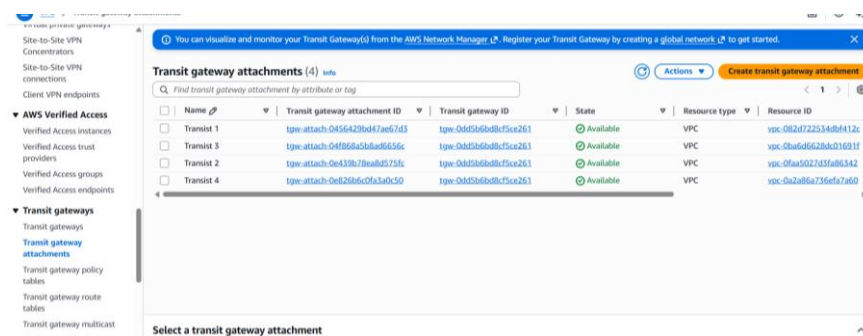
Name: My-Transit-Gateway



Step 10: Attach VPC-1 to Transit Gateway

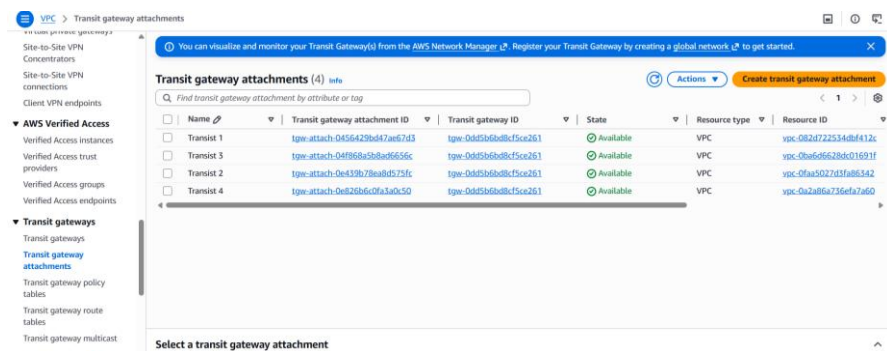
Transit Gateway Attachments → Create attachment

Select VPC-A and subnet



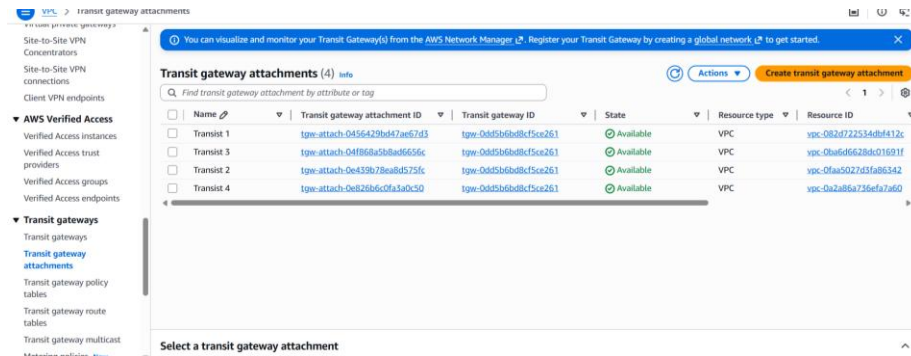
Step 11: Attach VPC-2 to Transit Gateway

Repeat same process



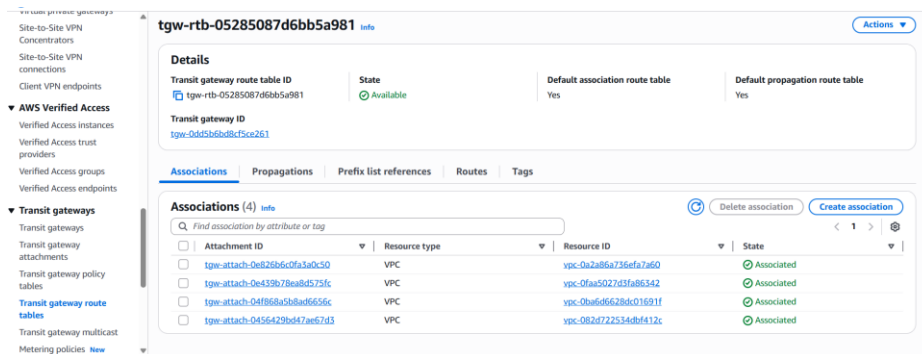
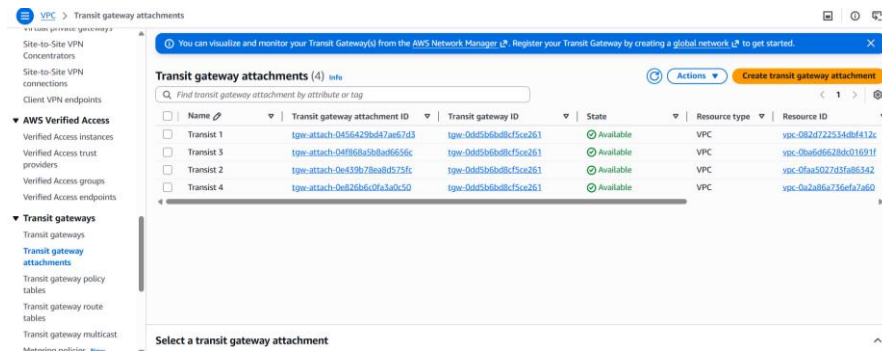
Step 12: Attach VPC-3 to Transit Gateway

Repeat same process



Step 13: Attach VPC-4 to Transit Gateway

Repeat same process

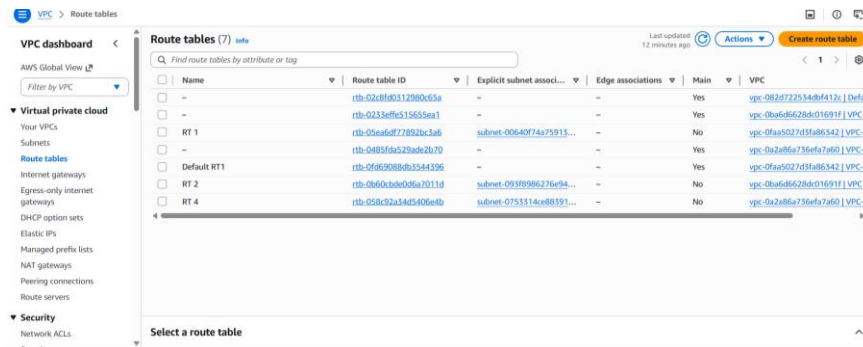


Step 14: Update Route Table of VPC-1

Add routes:

192.168.2.0/24 → Transit Gateway

192.168.3.0/24 → Transit Gateway

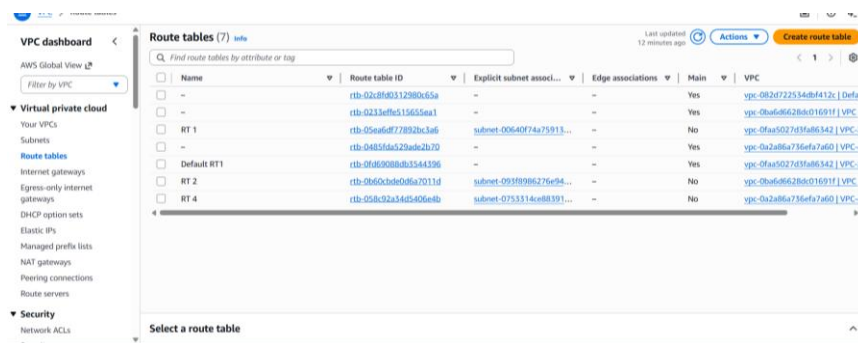


Step 15: Update Route Table of VPC-2

Add routes:

10.0.0.0/28 → Transit Gateway

10.0.0.0/28 → Transit Gateway

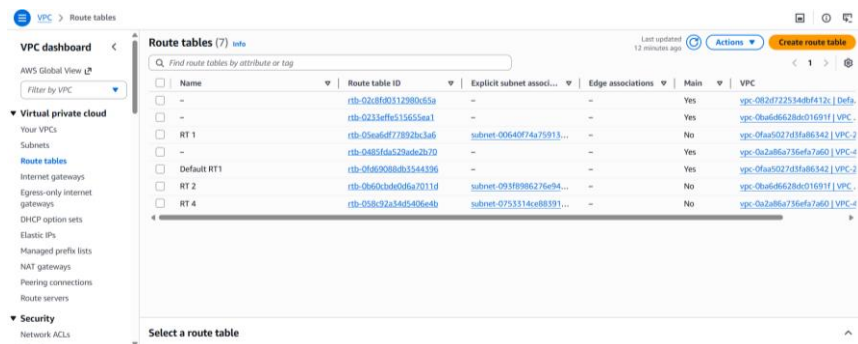


Step 16: Update Route Table of VPC-3

Add routes:

10.0.1.0/28 → Transit Gateway

10.0.1.0/28 → Transit Gateway



Step 17: Update Route Table of VPC-3

Add routes:

10.0.2.0/28 → Transit Gateway

10.0.2.0/28/24 → Transit Gateway

VPC dashboard < **Route tables (7)** info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-02c8f00312980d5a	-	-	Yes	vpc-082d722554dbf412c Defa...
-	rtb-0233effe515655ea1	-	-	Yes	vpc-0ba5d6628dc01691f VPC-
RT 1	rtb-02ea6d77892bc3a6	subnet-00640f74a75913...	-	No	vpc-0faa5027d3fa86342 VPC-2
-	rtb-0483fda529ade2b70	-	-	Yes	vpc-0a2a86a736efa7a60 VPC-4
Default RT1	rtb-0f869088db3544396	-	-	Yes	vpc-0faa5027d3fa86342 VPC-2
RT 2	rtb-0b60cb0d06a7011d	subnet-093f8986276e94...	-	No	vpc-0ba5d6628dc01691f VPC-
RT 4	rtb-058d92a34d5406e8b	subnet-0753314ce88391...	-	No	vpc-0a2a86a736efa7a60 VPC-4

Select a route table

Internet Gateway (IGW)

VPC dashboard < **Internet gateways (5)** info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
mumbai	igwe-00b6e7339765f1377	Detached	-	7080355287026
IGW Default	igwe-03eaa528a1028b4c1	Attached	vpc-082d722554dbf412c Default VPC1	7080355287026
IGW 4	igwe-02b14e0fc2287487	Attached	vpc-0a2a86a736efa7a60 VPC-4	7080355287026
IGW 3	igwe-0912c75633486598d	Attached	vpc-0ba5d6628dc01691f VPC-3	7080355287026
IGW2	igwe-0f0bb87423622efcc	Attached	vpc-0faa5027d3fa86342 VPC-2	7080355287026

Select an internet gateway above

Step 18: Launch EC2 instances in each VPC

Step 19: Test connectivity using ping between EC2 private IP addresses

```
ec2-user@ip-10-0-0-8 ~$ ping 10.0.2.8
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data.
64 bytes from 10.0.2.8: icmp_seq=1 ttl=126 time=1.07 ms
64 bytes from 10.0.2.8: icmp_seq=2 ttl=126 time=0.741 ms
64 bytes from 10.0.2.8: icmp_seq=3 ttl=126 time=1.60 ms
64 bytes from 10.0.2.8: icmp_seq=4 ttl=126 time=0.684 ms
64 bytes from 10.0.2.8: icmp_seq=5 ttl=126 time=0.772 ms
64 bytes from 10.0.2.8: icmp_seq=6 ttl=126 time=0.819 ms
^C
--- 10.0.2.8 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5124ms
rtt min/avg/max/mdev = 0.684/0.947/1.603/0.316 ms
ec2-user@ip-10-0-0-8 ~$
```



```

[ec2-user@ip-10-0-0-8 ~]$ ping 172.31.15.63
PING 172.31.15.63 (172.31.15.63) 56(84) bytes of data.
^C
--- 172.31.15.63 ping statistics ---
7 packets transmitted, 0 received, 100% packet loss, time 6241ms

[ec2-user@ip-10-0-0-8 ~]$ ping 10.0.0.8
PING 10.0.0.8 (10.0.0.8) 56(84) bytes of data.
64 bytes from 10.0.0.8: icmp_seq=1 ttl=127 time=0.020 ms
64 bytes from 10.0.0.8: icmp_seq=2 ttl=127 time=0.026 ms
64 bytes from 10.0.0.8: icmp_seq=3 ttl=127 time=0.027 ms
64 bytes from 10.0.0.8: icmp_seq=4 ttl=127 time=0.015 ms
^C
--- 10.0.0.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3155ms
rtt min/avg/max/mdev = 0.015/0.022/0.027/0.004 ms
[ec2-user@ip-10-0-0-8 ~]$ ping 10.0.1.10
PING 10.0.1.10 (10.0.1.10) 56(84) bytes of data.
64 bytes from 10.0.1.10: icmp_seq=1 ttl=126 time=1.42 ms
64 bytes from 10.0.1.10: icmp_seq=2 ttl=126 time=0.616 ms
64 bytes from 10.0.1.10: icmp_seq=3 ttl=126 time=0.700 ms
^C
--- 10.0.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2077ms
rtt min/avg/max/mdev = 0.616/0.911/1.418/0.359 ms
[ec2-user@ip-10-0-0-8 ~]$ ping 10.0.2.8
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data.
64 bytes from 10.0.2.8: icmp_seq=1 ttl=126 time=0.891 ms
64 bytes from 10.0.2.8: icmp_seq=2 ttl=126 time=0.885 ms
64 bytes from 10.0.2.8: icmp_seq=3 ttl=126 time=0.510 ms
^C
--- 10.0.2.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2018ms
rtt min/avg/max/mdev = 0.510/0.762/0.891/0.178 ms
[ec2-user@ip-10-0-0-8 ~]$

```

```

[ec2-user@ip-10-0-0-8 ~]$ ping 172.31.15.63
PING 172.31.15.63 (172.31.15.63) 56(84) bytes of data.
^C
--- 172.31.15.63 ping statistics ---
7 packets transmitted, 0 received, 100% packet loss, time 6241ms

[ec2-user@ip-10-0-0-8 ~]$ ping 10.0.0.8
PING 10.0.0.8 (10.0.0.8) 56(84) bytes of data.
64 bytes from 10.0.0.8: icmp_seq=1 ttl=127 time=0.020 ms
64 bytes from 10.0.0.8: icmp_seq=2 ttl=127 time=0.026 ms
64 bytes from 10.0.0.8: icmp_seq=3 ttl=127 time=0.027 ms
64 bytes from 10.0.0.8: icmp_seq=4 ttl=127 time=0.015 ms
^C
--- 10.0.0.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3155ms
rtt min/avg/max/mdev = 0.015/0.022/0.027/0.004 ms
[ec2-user@ip-10-0-0-8 ~]$ ping 10.0.1.10
PING 10.0.1.10 (10.0.1.10) 56(84) bytes of data.
64 bytes from 10.0.1.10: icmp_seq=1 ttl=126 time=1.42 ms
64 bytes from 10.0.1.10: icmp_seq=2 ttl=126 time=0.616 ms
64 bytes from 10.0.1.10: icmp_seq=3 ttl=126 time=0.700 ms
^C
--- 10.0.1.10 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2077ms
rtt min/avg/max/mdev = 0.616/0.911/1.418/0.359 ms
[ec2-user@ip-10-0-0-8 ~]$ ping 10.0.2.8
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data.
64 bytes from 10.0.2.8: icmp_seq=1 ttl=126 time=0.891 ms
64 bytes from 10.0.2.8: icmp_seq=2 ttl=126 time=0.885 ms
64 bytes from 10.0.2.8: icmp_seq=3 ttl=126 time=0.510 ms
^C
--- 10.0.2.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2018ms
rtt min/avg/max/mdev = 0.510/0.762/0.891/0.178 ms
[ec2-user@ip-10-0-0-8 ~]$

```



```

Amazon Linux 2023
~ ~ ~ #####
~ ~ ~ ####|
~ ~ ~ #/
~ ~ ~ V' -> https://aws.amazon.com/linux/amazon-linux-2023
~ ~ ~ /m/' ->
[ec2-user@ip-10-0-2-8 ~]$ ping 172.31.15.63
PING 172.31.15.63 (172.31.15.63) 56(84) bytes of data.
AC
--- 172.31.15.63 ping statistics ---
6 packets transmitted, 0 received, 100% packet loss, time 5205ms

[ec2-user@ip-10-0-2-8 ~]$ ping 10.0.0.8
PING 10.0.0.8 (10.0.0.8) 56(84) bytes of data.
64 bytes from 10.0.0.8: icmp_seq=1 ttl=126 time=0.836 ms
64 bytes from 10.0.0.8: icmp_seq=2 ttl=126 time=0.509 ms
64 bytes from 10.0.0.8: icmp_seq=3 ttl=126 time=0.652 ms
AC
--- 10.0.0.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2078ms
rtt min/avg/max/mdev = 0.509/0.665/0.836/0.133 ms
[ec2-user@ip-10-0-2-8 ~]$ exit
logout
Connection to 3.110.181.173 closed.
```

```
Warning: Permanently added '65.0.3.90' (ED25519) to the list of known hosts.
```

```
#  
##### Amazon Linux 2023  
#####  
#/  
V~'->  
  
m/-  
-/_/  
_/\_  
_/m/'
```

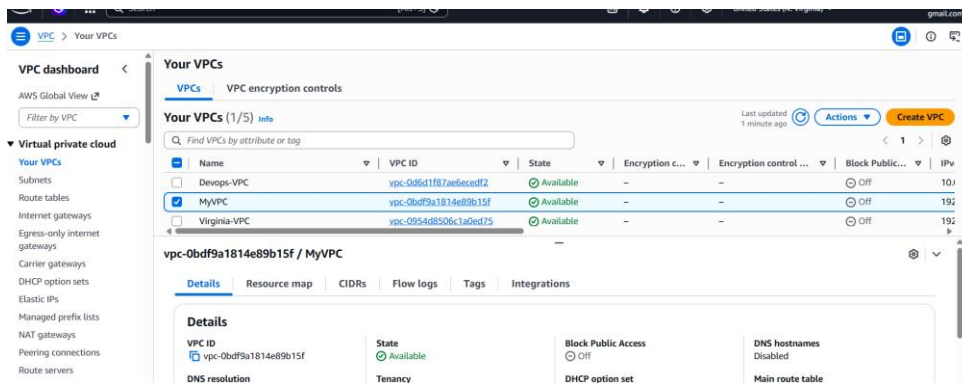
```
[ec2-user@ip-10-0-1-10 ~]$ ping 10.0.2.8  
PING 10.0.2.8 (10.0.2.8) 56(84) bytes of data:  
64 bytes from 10.0.2.8: icmp_seq=1 ttl=126 time=0.891 ms  
64 bytes from 10.0.2.8: icmp_seq=2 ttl=126 time=0.645 ms  
64 bytes from 10.0.2.8: icmp_seq=3 ttl=126 time=0.665 ms  
AC  
--- 10.0.2.8 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2046ms  
rtt min/avg/max/mdev = 0.645/0.733/0.891/0.111 ms  
[ec2-user@ip-10-0-1-10 ~]$
```

2) AWS Bastion Server and VPC Endpoint

Configuring Private AWS Service Access Using VPC Endpoints

STEP 1: Create VPC

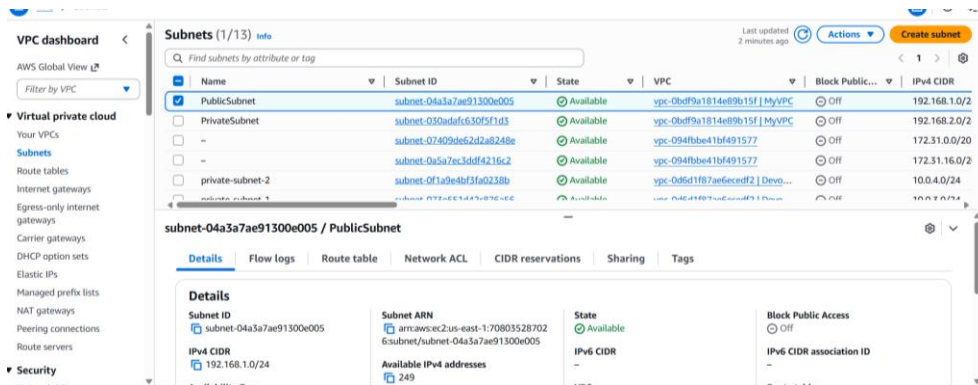
Name: MyVPC



STEP 2: Create Public Subnet

Name: Public-Subnet

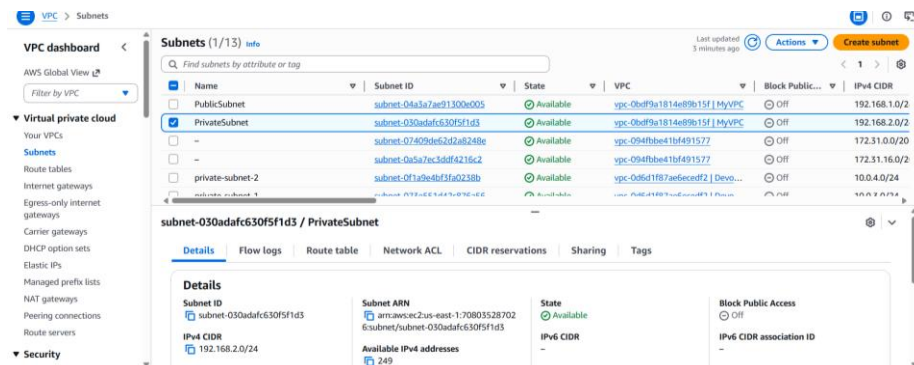
CIDR: 192.168.0.0/24



STEP 3: Create Private Subnet

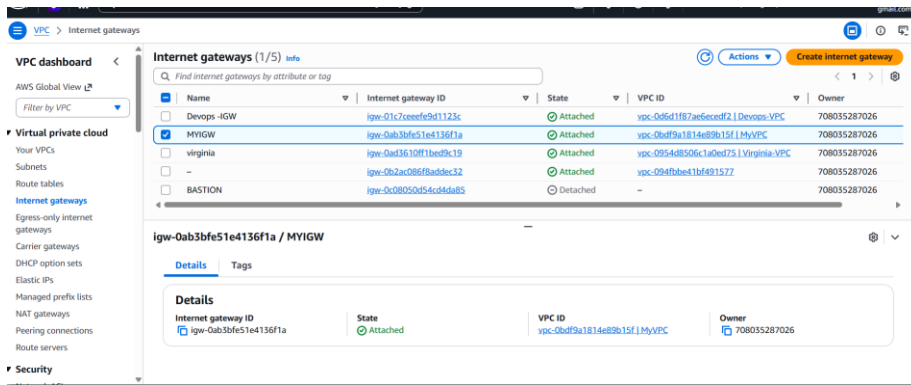
Name: Private-Subnet

CIDR: 192.168.1.0/24

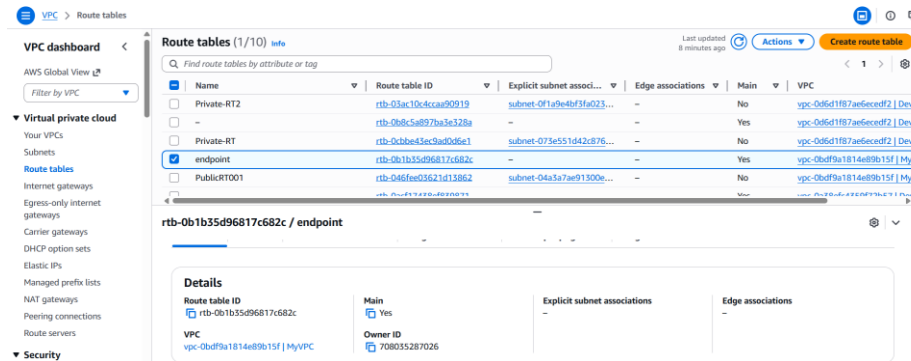


STEP 4: Create Internet Gateway

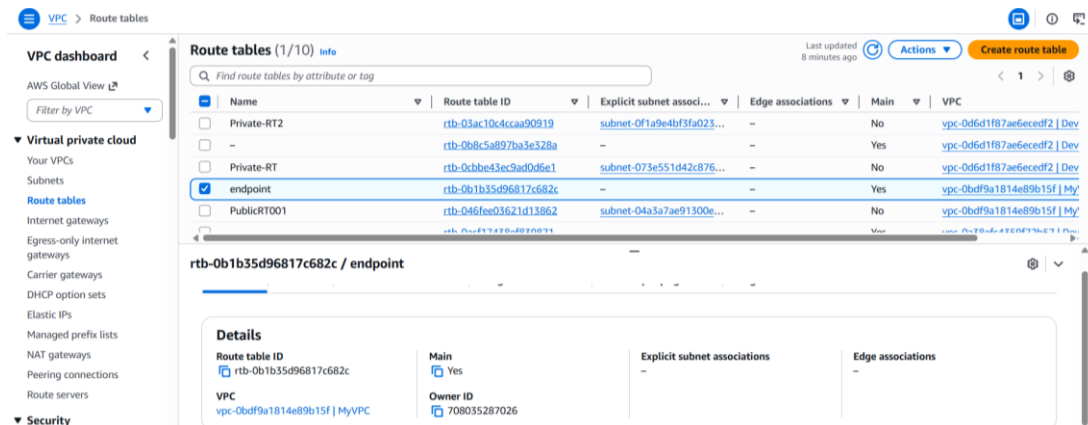
Attach Internet Gateway to MyVPC



STEP 5: Create Public Route Table
 Add route: 0.0.0.0/0 → Internet Gateway
 Associate Public Subnet



STEP 6: Create Private Route Table
 Associate Private Subnet
 Do not add Internet Gateway



VPC > Endpoints

Route servers

▼ Security

- Network ACLs
- Security groups

▼ PrivateLink and Lattice

- Getting started
- Endpoints
- Endpoint services
- Service networks
- Lattice services
- Resource configurations
- Resource gateways

Endpoints (1/1) Info

Find endpoints by attribute or tag

☒	Name	VPC endpoint ID	Endpoint type	Status	Service name
☒	MY ENDPOINT	vpce-0f0f4d198d31d71e4	Gateway	Available	com.amazonaws.us-east-

vpce-0f0f4d198d31d71e4 / MY ENDPOINT

Details Route tables Policy Tags

VPC > Route tables > rtb-0b1b35d96817c682c

PrivateLink and Lattice

- Getting started
- Endpoints
- Endpoint services
- Service networks
- Lattice services
- Resource configurations
- Resource gateways
- Target groups
- Domain verifications New

DNS firewall

- Rule groups
- Domain lists

Network Firewall

- Firewalls
- Firewall policies
- Network Firewall rule groups
- TLS inspection

rtb-0b1b35d96817c682c / endpoint

Details Info

Route table ID
rtb-0b1b35d96817c682c

VPC
vpc-0bdf9a1814e89b15f | MyVPC

Main
☒ Yes

Owner ID
708035287026

Explicit subnet associations
-

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Routes (3)

Filter routes

Destination	Target	Status	Propagated	Route Origin
pl-63a5400a	vpce-0f0f4d198d31d71e4	Active	No	Create Route
0.0.0.0/0	igw-0ab3bfe51e4136f1a	Active	No	Create Route
192.168.0.0/16	local	Active	No	Create Route Table

CloudShell Feedback Console Mobile App

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STEP 7: Launch Bastion Server

Launch EC2 in Public Subnet

Enable Public IP

EC2 > Instances

Dashboard

AWS Global View

Events

Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager New

Images

- AMIs
- AMI Catalog

Elastic Block Store

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

Running

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	endpoint	i-04f0ee25389afea79	Running	t2.micro	Initializing	View alarms +	us-east-1d	-
☒	BASTION	i-05a5591152f65569d	Running	t2.micro	Initializing	View alarms +	us-east-1d	-

i-05a5591152f65569d (BASTION)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID
i-05a5591152f65569d

IPv6 address
-

Hostname type
IP name: ip-192-168-1-105.ec2.internal

Public IPv4 address
54.85.96.15 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-192-168-1-105.ec2.internal

Private IPv4 addresses
192.168.1.105

Public DNS
-

Disable Public IP

[illegible]

